

Name: Octavio Morales Date: November 6, 2022

Point values are assigned for each question.

Points earned: ____ / 100

1. Consider the algorithm on page 148 in the textbook for Binary Reflected Gray Codes. What change(s) would you make so that it generates the binary numbers **in order** for a given length n ? Your algorithm must be recursive and keep the same structure as the one in the textbook. Describe only the change(s). (10 points)

Most algorithms BRGC(n) would stay the same. However, in the else statement, instead of copying list one to list two in reverse order, we would copy it in the same order. "copy list L1 to list L2 in reversed order" would become "copy list L1 to list L2 in **the same order**"

2. Show the steps to multiply 72×93 with Russian peasant multiplication, as seen in Figure 4.11b on page 154 in the textbook. (10 points)

72	93
36	186
18	372
9	744
4	1488
2	2976
1	5952

$$744 + 5952 = 6696$$

$$72 \times 93 = 6696$$

3. Suppose you use the LomutoPartition() function on page 159 in the textbook in your implementation of Quicksort. (10 points, 5 points each)

a) Describe the types of input that cause Quicksort to perform its worst-case running time.

Quicksort would perform its worst-case running time in the LomutoPartition() function when the sequence of numbers is sorted in reverse order.

b) What is that worst-case running time?

When this happens, the worst-case running time would be $\Theta(n^2)$.

4. Compute 2205×1132 by applying the divide-and-conquer Karatsuba algorithm outlined in the text. Repeat the process until the numbers being multiplied are each 1 digit. For each multiplication, show the values of c_2 , c_1 , and c_0 . Do not skip steps. (10 points)

4. $2205 \cdot 1132 = 2496060$

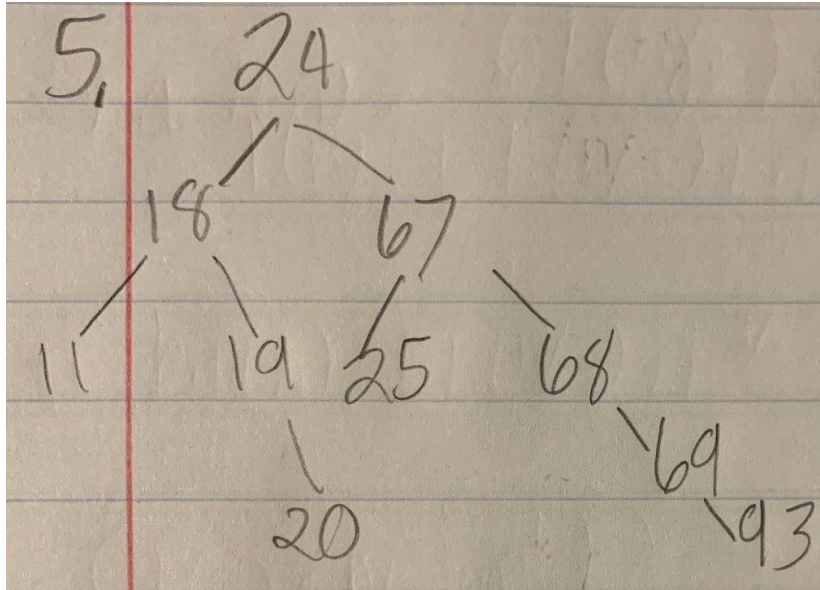
$a_1 = 22$ $b_1 = 11$ $c_2 = a_1 \cdot b_1 = 22 \cdot 11 = 242$
 $a_0 = 5$ $b_0 = 32$ $c_0 = a_0 \cdot b_0 = 5 \cdot 32 = 160$
 $c_1 = (a_1 + a_0) \cdot (b_1 + b_0) - (c_2 + c_0) = (22 + 5) \cdot (11 + 32) - (242 + 160)$
 $= 27 \cdot 43 - 402 = 1161 - 402 = 759$

$22 \cdot 11 = 242$
 $a_1 = 2$ $b_1 = 1$ $c_2 = a_1 \cdot b_1 = 2 \cdot 1 = 2$
 $a_0 = 2$ $b_0 = 1$ $c_0 = a_0 \cdot b_0 = 2 \cdot 1 = 2$
 $c_1 = (a_1 + a_0) \cdot (b_1 + b_0) - (c_2 + c_0) = (2 + 2) \cdot (1 + 1) - (2 + 2)$
 $= 4$

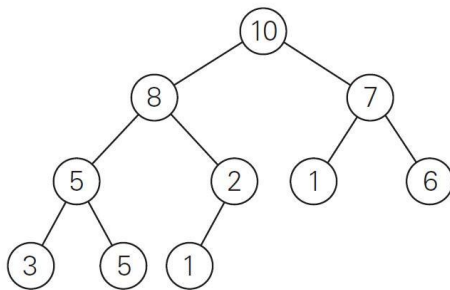
$5 \cdot 32$
 $a_1 = 0$ $b_1 = 3$ $c_2 = a_1 \cdot b_1 = 0 \cdot 3 = 0$
 $a_0 = 5$ $b_0 = 2$ $c_0 = a_0 \cdot b_0 = 5 \cdot 2 = 10$
 $c_1 = (a_1 + a_0) \cdot (b_1 + b_0) - (c_2 + c_0) = (0 + 5) \cdot (3 + 2) - (0 + 10)$
 $= 5 \cdot 5 - 10 = 15$

$27 \cdot 43$
 $a_1 = 2$ $b_1 = 4$ $c_2 = a_1 \cdot b_1 = 2 \cdot 4 = 8$
 $a_0 = 7$ $b_0 = 3$ $c_0 = a_0 \cdot b_0 = 7 \cdot 3 = 21$
 $c_1 = (a_1 + a_0) \cdot (b_1 + b_0) - (c_2 + c_0) = (2 + 7) \cdot (4 + 3) - (8 + 21) = 9 \cdot 7 - 29 = 63 - 29 = 34$

5. Draw the binary search tree after inserting the following keys: 24 18 67 68 69 25 19 20 11 93 (10 points)



6. Consider the following binary tree. (16 points, 2 points each)



- Traverse the tree preorder.
10, 8, 5, 3, 5, 2, 1, 7, 1, 6
- Traverse the tree inorder.
3, 5, 5, 8, 1, 2, 10, 1, 7, 6
- Traverse the tree postorder.
3, 5, 5, 1, 2, 8, 1, 6, 7, 10
- How many internal nodes are there?
5 internal nodes: 10, 8, 7, 5, 2
- How many leaves are there?
5 leaves: 3, 5, 1, 1, 6
- What is the maximum width of the tree?
The maximum width of this tree is 4 (composed of 5, 2, 1, 6)
- What is the height of the tree?
The height of the tree is 3.
- What is the diameter of the tree?
The diameter of the tree is 5.

7. Use the Master Theorem to give tight asymptotic bounds for the following recurrences. (25 points, 5 points each)

CS 385, Homework 4: Decrease/Divide and Conquer

7. a. $T(n) = 2T(n/4) + 1$ $a=2$ $b^k = 4^0 = 1$ $2 > 1$
 $a > b^k$
 $T(n) \in O(n^{\log_4 2}) \rightarrow T(n) \in O(n^{1/2})$

b. $T(n) = 2T(n/4) + \sqrt{n}$ $a=2$ $b^k = 4^{1/2} = 2$ $2 = 2$
 $a = b^k$
 $T(n) \in O(\sqrt{n} \cdot \log n)$

c. $T(n) = 2T(n/4) + n$ $a=2$ $b^k = 4^1 = 4$ $2 < 4$
 $a < b^k$
 $T(n) \in O(n)$

d. $T(n) = 2T(n/4) + n^2$ $a=2$ $b^k = 4^2 = 16$ $2 < 16$
 $a < b^k$
 $T(n) \in O(n^2)$

e. $T(n) = 2T(n/4) + n^3$ $a=2$ $b^k = 4^3 = 64$ $2 < 64$
 $a < b^k$
 $T(n) \in O(n^3)$

8. Consider the following function. (9 points)

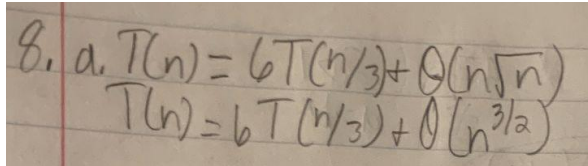
```
int function(int n) {
    if(n <= 1) {
        return 0;
    }
    int temp = 0;
    for(int i = 1; i <= 6; i++) {
        temp += function(n / 3);
    }
    for(int i = 1; i <= n; i++) {
```

```

        for(int j = 1; j * j <= n; j++) {
            temp++;
        }
    }
    return temp;
}

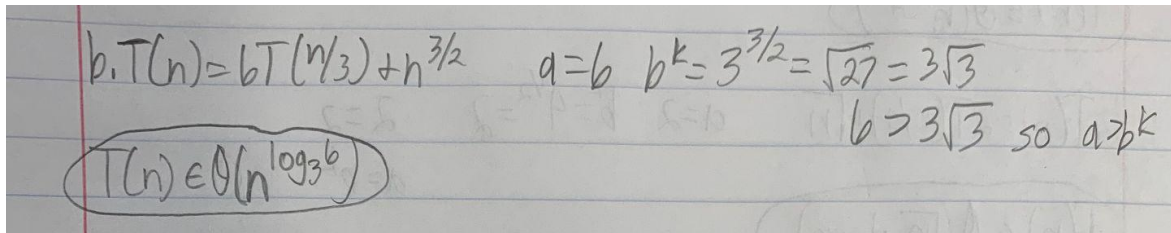
```

a) Write an expression for the runtime $T(n)$ for the function. (4 points)



8. a. $T(n) = 6T(n/3) + O(n\sqrt{n})$
 $T(n) = 6T(n/3) + O(n^{3/2})$

b) Use the Master Theorem to give a tight asymptotic bound. Simplify your answer as much as possible. (5 points)



b. $T(n) = 6T(n/3) + n^{3/2}$ $a=6$ $b^k = 3^{3/2} = \sqrt{27} = 3\sqrt{3}$
 $6 > 3\sqrt{3}$ so $a > b^k$
 $T(n) \in O(n^{\log_3 6})$

I pledge my honor that I have abided by the Stevens Honor System. Octavio Morales